

Exp: 3 Performance Analysis of IIR Filters

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

Program:

```
clc; clear; close all;

% Specifications

Fs=2000; Rp=1; Rs=40;

Wp=[300 600]/(Fs/2);

Ws=[200 700]/(Fs/2);

% Butterworth Filter

[n1,W1]=buttord(Wp,Ws,Rp,Rs);

[b1,a1]=butter(n1,W1,'bandpass');

% Chebyshev Type-I Filter

[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);

[b2,a2]=cheby1(n2,Rp,W2,'bandpass');

% Magnitude Response

figure;

freqz(b1,a1); title('Butterworth Filter');

figure;

freqz(b2,a2); title('Chebyshev Type-I Filter');

% Group Delay

figure;

grpdelay(b1,a1); hold on;

grpdelay(b2,a2);

legend('Butterworth','Chebyshev-I');
```

% Pole-Zero Plot

figure;

subplot(121),zplane(b1,a1),title('Butterworth')

subplot(122),zplane(b2,a2),title('Chebyshev-I')

% Impulse Response

figure;

subplot(211),impz(b1,a1),title('Butterworth Impulse')

subplot(212),impz(b2,a2),title('Chebyshev Impulse')

% Pole Magnitude

figure;

subplot(121),stem(abs(roots(a1)),'filled')

title('Butterworth Pole Magnitude')

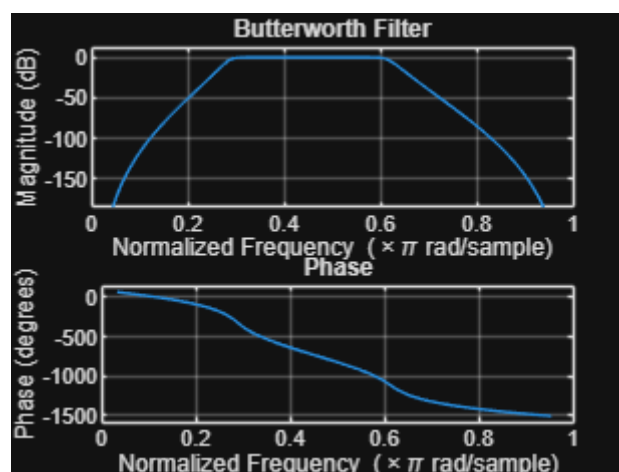
subplot(122),stem(abs(roots(a2)),'filled')

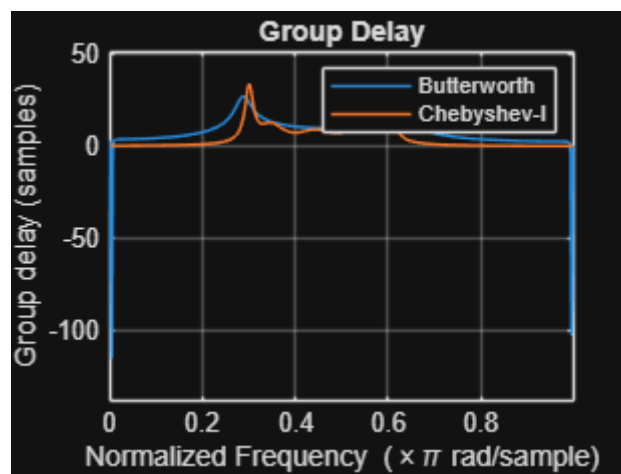
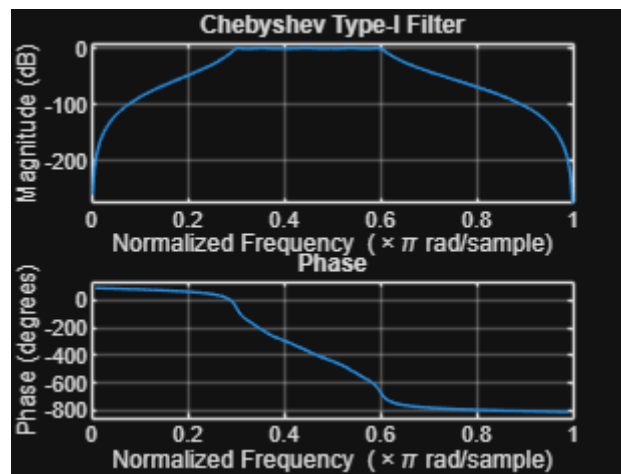
title('Chebyshev Pole Magnitude')

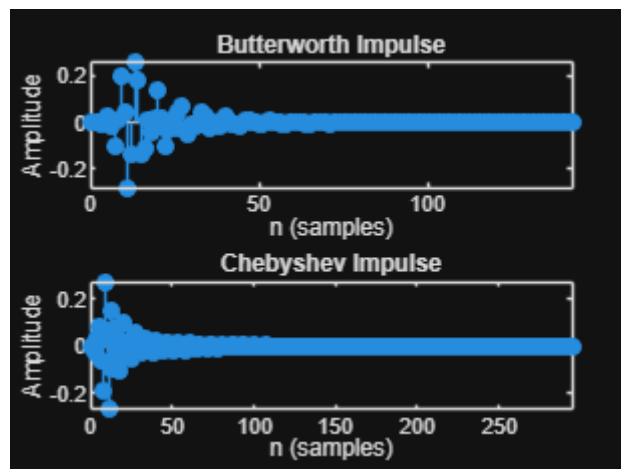
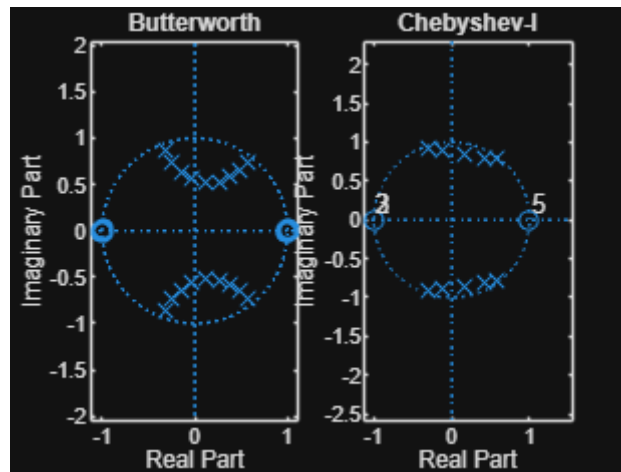
disp(['Butterworth Order = ',num2str(n1)])

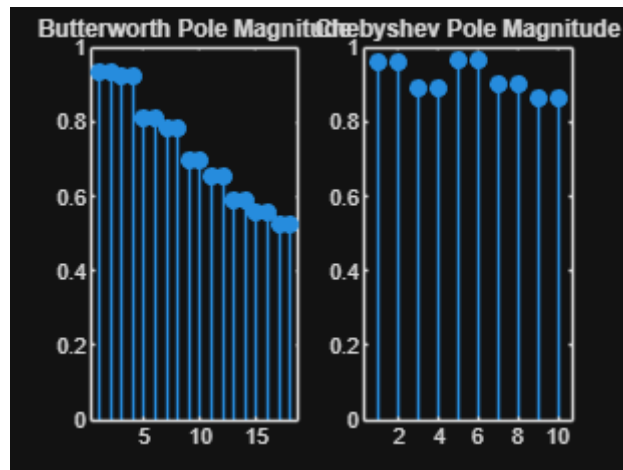
disp(['Chebyshev Order = ',num2str(n2)])

Outputs:









Butterworth Order = 9

Chebyshev Order = 5